

# Digital Transformation – Core Concepts

Unit 4 – Lecture 1

# Learning Objectives

- Define digital transformation
- Differentiate digitization, digitalization, and digital transformation
- Understand why digital transformation is important
- Identify real-world examples

# What is Digital Transformation?

- Digital transformation is the **use of digital technologies** to fundamentally change **how** organizations **operate** and **deliver** value
- It affects:
  - a. **Business processes**
  - b. **Business models**
  - c. **Customer experience**
- It is **not only about technology**, but also strategy and culture
- **Key idea:**  
*“Doing things differently using digital technologies”*

# Example - School

## **Before** (Old way)

- Teacher writes notes on the blackboard
- Students copy in notebooks
- Homework is checked manually

## **After** (Digital transformation)

- Teacher uses a projector or smart board
- Notes are shared on WhatsApp / Google Classroom
- Homework is submitted online
- Same school, but **doing things differently using technology.**

# Example 2: Shop / Grocery Store

## Before

- You go to the shop
- Ask the shopkeeper
- Pay with cash

## After

- You order from an app
- Pay using QR code or mobile wallet
- Items are delivered to your home
- The shop still sells things, but **in a new digital way.**

# Example 3: Food

## **Before**

- Go to a restaurant
- Wait for food
- Pay at the counter

## **After**

- Order food from an app
- Track your order
- Food comes home

Better and faster experience for customers.

# Example 4: Games

## Before

- Play games outside only
- Board games like Ludo

## After

- Play online games
- Play with friends from different places
- Save progress digitally

Same fun, but **new digital method.**

# Evolution Toward Digital Transformation

- **Manual** systems → **Computerized** systems (e.g., ledgers → Excel)
- **Standalone** software → **Internet-based** systems
- **Traditional** economy → **Digital** economy
- **Physical** services → **Digital** platforms
- **Example:**
- Traditional bank → Online banking → Mobile banking app



# Digitization

- Conversion of **analog information** into **digital form**
- Focus: **Data**
- No major process change
- **Examples:**
  - Scanning paper documents into PDF
  - Converting handwritten records into Excel files
  - Government offices scanning land ownership documents.

*Digitization = Data becomes digital*

# Digitalization

- Use of digital technologies to **improve existing processes**
- Focus: **Efficiency**
- Processes remain mostly the same
- **Examples:**
  - Online form instead of paper form
  - Email instead of postal letters
  - *Digitalization = **Process improvement***

# Digital Transformation (compared)

- Complete **rethinking of processes and models**
- Focus: **Innovation & value creation**
- Changes how organizations work and compete
- **Examples:**
  - Ride-hailing apps replacing traditional taxis
  - Online education platforms replacing physical classrooms
  - *Digital Transformation = **Business change***

# Comparison Table

| Aspect  | Digitization  | Digitalization | Digital Transformation |
|---------|---------------|----------------|------------------------|
| Focus   | Data          | Process        | Business model         |
| Scope   | Narrow        | Medium         | Broad                  |
| Impact  | Low           | Moderate       | High                   |
| Example | Scanned files | Online forms   | Digital platforms      |

# Why Digital Transformation is Important

- Rising customer expectations
- Global competition
- Cost efficiency
- Innovation and growth
- Survival in digital economy
- Idea:  
“Organizations that don’t transform digitally risk becoming irrelevant.”

# Real-World Examples

- Banking:
  - Mobile banking, QR payments
- Education:
  - Online classes, LMS
- Government:
  - E-governance services
- **Nepal Examples:**
  - eSewa, Khalti
  - Online exam forms
  - Nagarik App

# Real-World Failures

- **Kodak – Photography**
- **What happened:**
  - + Kodak invented digital cameras **but stuck to selling film**
  - + They **ignored** the **shift to** digital photography
- **Result:**
  - + Lost market share to Canon, Nikon, and smartphone cameras
- **Lesson:**
  - + Ignoring digital change can make even a giant company disappear



# Real-World Failures

- **Nokia – Mobile Phones**
- **What happened:**
  - + Nokia dominated phones but **ignored smartphones and app ecosystems**
  - + Competitors like Apple (iPhone) and Samsung took over
- **Result:**
  - + **Lost market leadership**
- **Lesson:**
  - + Not innovating with technology can make you irrelevant





# Real-World Failures

- **Yahoo** – Internet Services
- **What happened:**
  - + Yahoo **did not innovate and compete** with Google and Facebook
  - + **Missed opportunities** in search, social media, and mobile
- **Result:**
  - + Lost relevance; sold parts of the company
- **Lesson:**
  - + Even tech companies must keep transforming



# Real-World Failures

- **Toys “R” Us – Retail**
- **What happened:**
  - + Didn’t build strong **online shopping presence**
  - + Amazon and Walmart captured the market
- **Result:**
  - + Filed for bankruptcy in 2017
- **Lesson:**
  - + Digital transformation in retail is necessary to survive



# Digital Transformation in Digital Economy

- **Foundation of digital economy**

- **Enables:**

- Platform businesses
- Digital payments
- Global digital services

- **Value comes from:**

- Data
- Networks
- Technology

# Key Takeaways

- Digital transformation is **more than technology**
- Clear difference between **digitization**, **digitalization**, and **transformation**
- Digital transformation **drives** the digital economy
- Real-world **impact** is **visible everywhere**

# Discussion

- Is Facebook an example of digitalization or digital transformation?
- Why is digital transformation necessary today?